

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CS A5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I K.Dhilip/192365013 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 5	/ 6	/ 4	/ 3	/ 2	
2	/ 5	/ 6	/ 4	/ 3	/ 2	
3	/ 5	/ 6	/ 4	/ 3	/ 2	
4	/ 5	/ 6	/ 4	/ 3	/ 2	
5	/ 5	/ 6	/ 4	/ 3	/ 2	
OVERALL RUBRICS VALUE						
	/ 5	/ 5	/ 4	/ 4	/ 2	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link

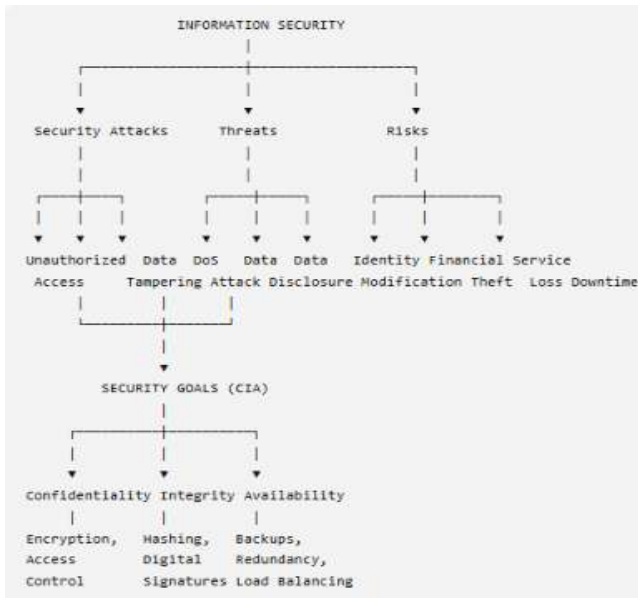


Table Showing the Links

Security Attack	Threat Caused	Risk	Security Goal	How the Goal Mitigates the Risk	Real-World Example
Unauthorized Access (Hacking/Phishing)	Disclosure of confidential information	Identity theft, financial loss, privacy breach	Confidentiality	Encryption, authentication, and access controls prevent unauthorized users from viewing sensitive data.	The Equifax data breach (2017) exposed personal information of about 147 million people after attackers gained unauthorized access.
Data Tampering (Modification Attack)	Alteration of data	Incorrect records, fraud, poor business decisions	Integrity	Hashing, digital signatures, and integrity checks detect and prevent unauthorized changes.	Attackers modifying bank transaction records would cause incorrect account balances and financial fraud.
Denial of Service (DoS/DDoS)	Service interruption	Website downtime, lost revenue, reduced customer trust	Availability	Redundant servers, backups, firewalls, and load balancing keep services available during attacks.	The Dyn DNS DDoS attack (2016) disrupted access to websites such as Twitter, Netflix, and Spotify for many users.

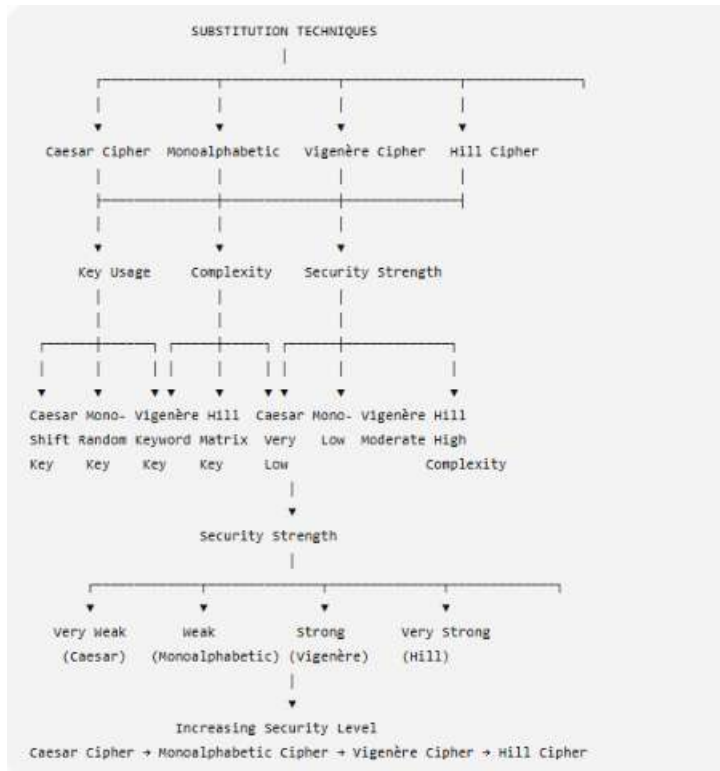
Summary

- Security Attacks are actions carried out by attackers.
- These attacks create Threats that exploit vulnerabilities.
- Threats result in Risks, causing damage to people or organizations.
- The CIA Security Goals reduce these risks:
 - Confidentiality protects information from unauthorized access.
 - Integrity ensures information remains accurate and unaltered.
 - Availability ensures systems and data remain accessible to authorized users.

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2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



Comparison Table

Technique	Key Usage	Complexity	Security Strength	Reason
Caesar Cipher	Single shift key (0–25)	Very Low	Very Weak	Only 25 possible keys; easily broken using brute force or frequency analysis.
Monoalphabetic Cipher	One fixed substitution alphabet	Low	Weak	Larger key space than Caesar but still vulnerable to frequency analysis.
Vigenère Cipher	Repeating keyword	Moderate	Strong	Uses multiple substitution alphabets, making frequency analysis more difficult.
Hill Cipher	Matrix key (2×2, 3×3, etc.)	High	Very Strong	Encrypts groups of letters using matrix multiplication, making statistical attacks much harder.

Analysis: Which Technique is Most Secure?

Hill Cipher is the most secure among these four techniques because:

- Uses matrix-based encryption instead of simple letter substitution.
- Encrypts blocks of letters rather than individual characters.
- Produces greater diffusion, so changing one letter affects multiple ciphertext characters.
- Much more resistant to frequency analysis than Caesar and Monoalphabetic ciphers.
- Larger key space due to matrix keys increases the difficulty of brute-force attacks.

Security Ranking

Rank	Technique	Security Level
1	Hill Cipher	Very Strong
2	Vigenère Cipher	Strong
3	Monoalphabetic Cipher	Weak
4	Caesar Cipher	Very Weak

Conclusion

Among the four substitution techniques:

Annexure A

- Caesar Cipher is the simplest and least secure.
- Monoalphabetic Cipher improves security but remains vulnerable to frequency analysis.
- Vigenère Cipher offers stronger security by using multiple substitution alphabets.
- Hill Cipher provides the highest security because it encrypts blocks of text using matrix operations, making it significantly more resistant to cryptanalysis.

Overall Security Order:

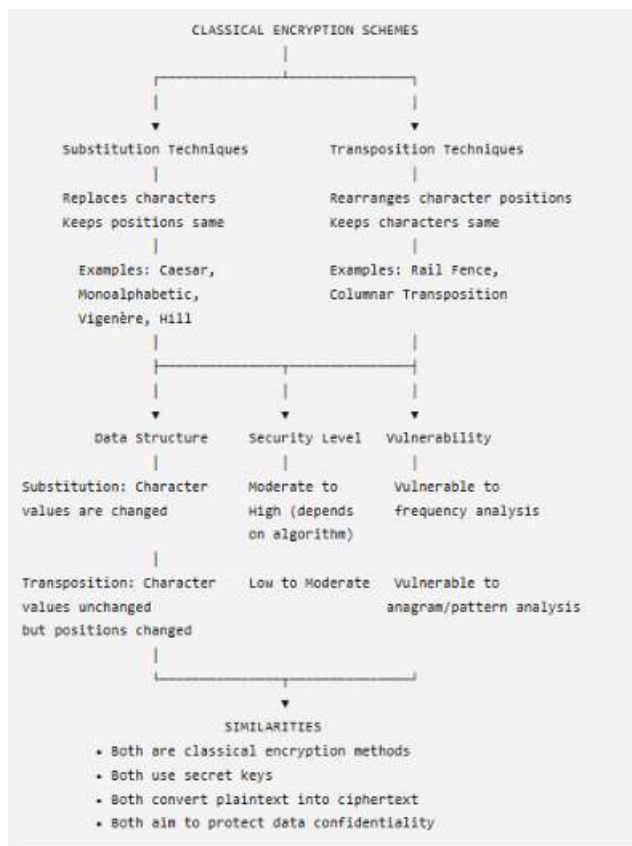
Hill Cipher > Vigenère Cipher > Monoalphabetic Cipher > Caesar Cipher

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why



(a) Differences and Similarities

Differences

Feature	Substitution Techniques	Transposition Techniques
Method	Replace characters with other characters	Rearrange the positions of characters
Character Values	Changed	Not changed
Character Positions	Remain the same	Changed
Examples	Caesar, Monoalphabetic, Vigenère, Hill	Rail Fence, Columnar Transposition
Main Principle	Substitution	Permutation

Similarities

- Both are classical encryption techniques.
- Both convert plaintext into ciphertext.
- Both use encryption keys.
- Both aim to ensure confidentiality of information.
- Both can be broken using cryptanalysis if used alone.

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(i) Effect on Data Structure and Security Level

Technique	Effect on Data Structure	Security Level
Substitution	Changes character values but preserves their order.	Moderate to High (depends on the algorithm used)
Transposition	Keeps character values the same but changes their positions.	Low to Moderate

(ii) Which Technique is More Vulnerable and Why?

Transposition Techniques are generally more vulnerable because:

- The original letters remain unchanged.
- An attacker can exploit language patterns and letter frequencies.
- Once the arrangement pattern (key) is discovered, the message can be reconstructed easily.

Substitution Techniques

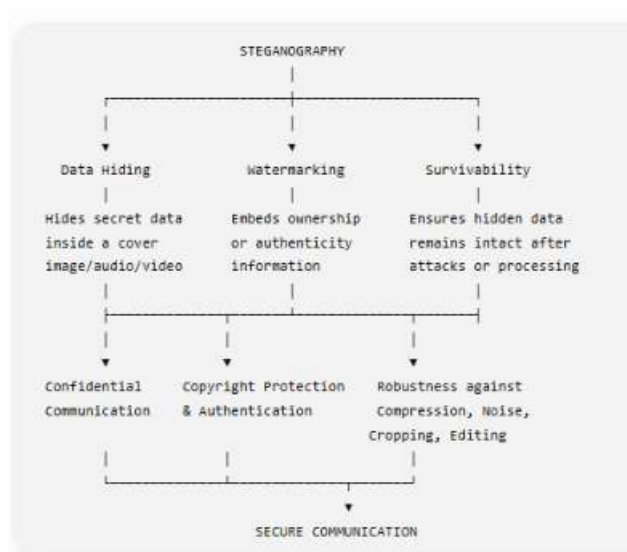
- Basic substitution methods (such as Caesar Cipher) are also vulnerable.
- Advanced substitution methods (such as Vigenère and Hill Cipher) provide stronger protection because they alter the character values and are harder to analyze.

Conclusion

- Substitution techniques change the actual characters, providing better protection than simple transposition methods.
- Transposition techniques only rearrange character positions, making them more susceptible to pattern-based cryptanalysis.
- In practice, modern cryptographic systems combine both substitution and transposition (e.g., in block ciphers like AES) to achieve much stronger security.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each



(a) Relationships Between the Concepts

Concept	Relationship
Steganography	The overall technique of hiding information within digital media.
Data Hiding	Conceals secret information so others are unaware that communication exists.
Watermarking	Embeds ownership or authenticity information into digital content.
Survivability	Ensures that the hidden data or watermark remains intact even after compression, editing, or attacks.

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(b) How Each Contributes to Secure Communication

Concept	Contribution to Secure Communication
Data Hiding	Protects confidentiality by concealing the existence of the secret message within an image, audio, or video file.
Watermarking	Protects authenticity and ownership by proving the source of digital content and discouraging unauthorized copying.
Survivability	Maintains the hidden information even after image compression, cropping, resizing, or noise, ensuring reliable communication.

(c) Real-World Applications

Concept	Real-World Application
Data Hiding	Military and intelligence agencies hide confidential messages inside digital images for covert communication.
Watermarking	Stock photo websites (e.g., Shutterstock or Getty Images) add digital watermarks to protect copyrighted images.
Survivability	Digital TV broadcasting and multimedia streaming use robust watermarks that remain detectable even after compression or format conversion.

Conclusion

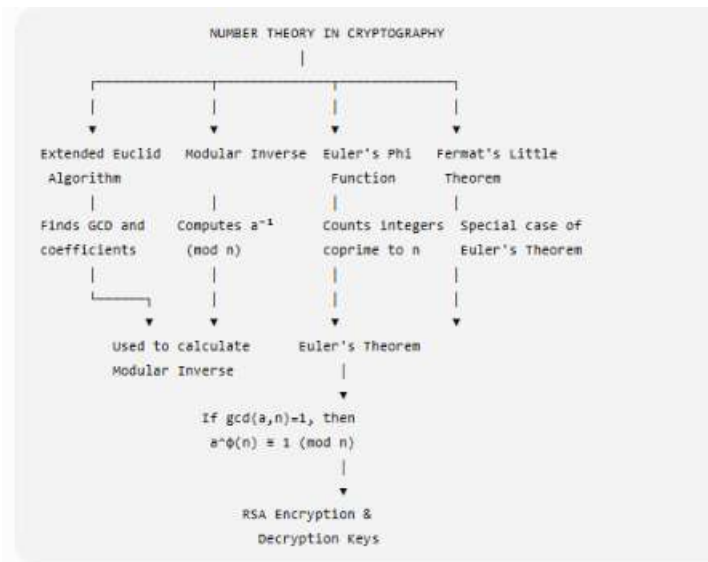
Steganography provides secure communication by combining three important concepts:

- Data Hiding → Conceals the existence of secret information (confidentiality).
- Watermarking → Protects ownership and authenticity (copyright protection).
- Survivability → Ensures the hidden information remains recoverable despite modifications (robustness).

Together, these techniques enable secure, reliable, and authentic transmission of digital information.

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify



(a) Mathematical Connections

Concept	Mathematical Connection
Extended Euclid Algorithm	Computes the GCD of two numbers and the coefficients needed to find the modular inverse.
Modular Inverse	Exists only if $\gcd(a, n) = 1$ and is calculated using the Extended Euclid Algorithm.
Euler's Phi Function ($\phi(n)$)	Gives the number of positive integers less than n that are relatively prime to n .
Euler's Theorem	Uses $\phi(n)$: $a^{\phi(n)} \equiv 1 \pmod{n}$ when $\gcd(a, n) = 1$.
Fermat's Little Theorem	A special case of Euler's Theorem for prime numbers: $a^{p-1} \equiv 1 \pmod{p}$.

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(b) Role in Encryption and Decryption

Concept	Role in Cryptography
Extended Euclid Algorithm	Computes the private key by finding the modular inverse in RSA.
Modular Inverse	Used to calculate the decryption key from the encryption key.
Euler's Phi Function	Determines RSA key generation by computing $\phi(n) = (p-1)(q-1)$.
Euler's Theorem	Forms the mathematical foundation that guarantees correct RSA encryption and decryption.
Fermat's Little Theorem	Used in modular exponentiation, primality testing, and optimization in cryptographic algorithms.

(c) Most Critical Concept for Public Key Cryptography

Euler's Theorem is the most critical concept.

Justification

- It provides the mathematical foundation of the RSA algorithm, the most widely used public-key cryptosystem.
- It guarantees that encryption and decryption are inverse operations:

$$M^{ed} \equiv M \pmod{n}$$

where:

- e = public key exponent
 - d = private key exponent
- Without Euler's Theorem, RSA cannot ensure that the original plaintext is recovered after decryption.
- The theorem also relies on Euler's Phi Function, making $\phi(n)$ an essential component of key generation.

Summary Table

Concept	Primary Purpose	Used In
Extended Euclid Algorithm	Finds GCD and modular inverse	RSA key generation
Modular Inverse	Computes private key	RSA decryption
Euler's Phi Function	Calculates $\phi(n)$ for key generation	RSA
Euler's Theorem	Mathematical basis for encryption/decryption	RSA (Core Principle)
Fermat's Little Theorem	Primality testing and modular arithmetic	RSA, Diffie-Hellman, ElGamal

Conclusion

The number theory concepts are closely connected:

- Extended Euclid Algorithm → Computes the Modular Inverse.
- Euler's Phi Function → Enables Euler's Theorem.
- Fermat's Little Theorem → A special case of Euler's Theorem for prime numbers.
- Together, these concepts enable secure public-key cryptography, particularly RSA.

Most Critical Concept: Euler's Theorem, because it provides the mathematical proof that makes RSA encryption and decryption work correctly.

Annexure A

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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